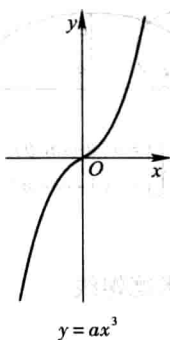
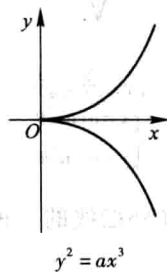


附录Ⅲ 几种常用的曲线

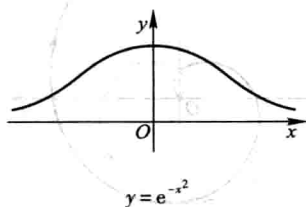
(1) 三次抛物线



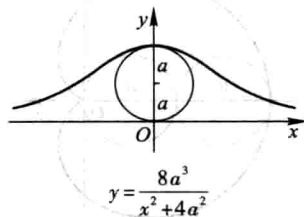
(2) 半立方抛物线



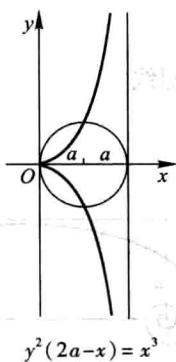
(3) 概率曲线



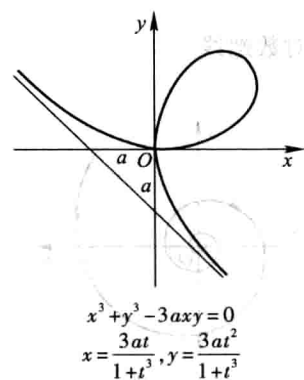
(4) 箕舌线



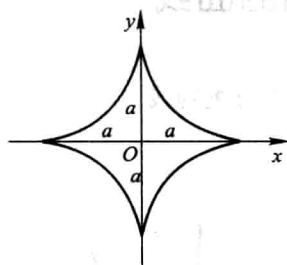
(5) 蔓叶线



(6) 笛卡儿叶形线

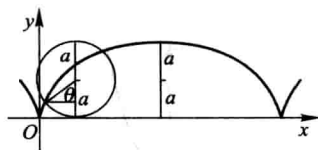


(7) 星形线(内摆线的一种)



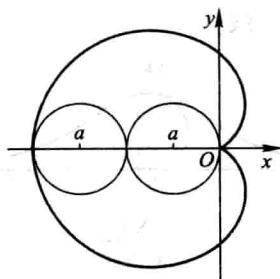
$$\begin{cases} x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \\ x = a \cos^3 \theta \\ y = a \sin^3 \theta \end{cases}$$

(8) 摆线



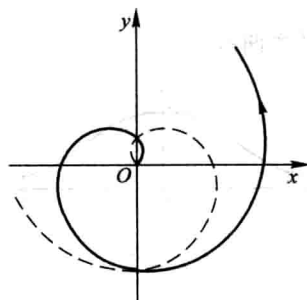
$$\begin{cases} x = a(\theta - \sin \theta) \\ y = a(1 - \cos \theta) \end{cases}$$

(9) 心形线(外摆线的一种)



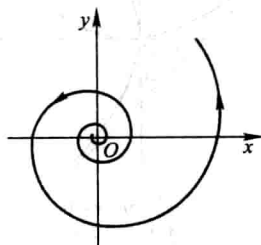
$$\begin{cases} x^2 + y^2 + ax = a \sqrt{x^2 + y^2} \\ \rho = a(1 - \cos \theta) \end{cases}$$

(10) 阿基米德螺线



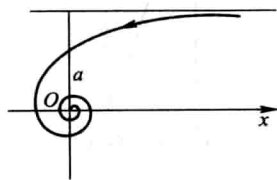
$$\rho = a\theta$$

(11) 对数螺线



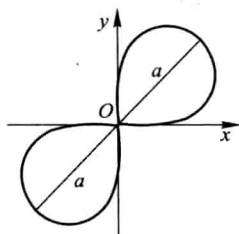
$$\rho = e^{a\theta}$$

(12) 双曲螺线



$$\rho\theta = a$$

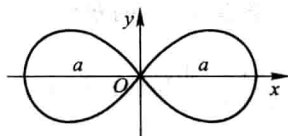
(13) 伯努利双纽线



$$(x^2 + y^2)^2 = 2a^2 xy$$

$$\rho^2 = a^2 \sin 2\theta$$

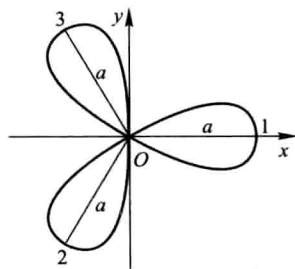
(14) 伯努利双纽线



$$(x^2 + y^2)^2 = a^2 (x^2 - y^2)$$

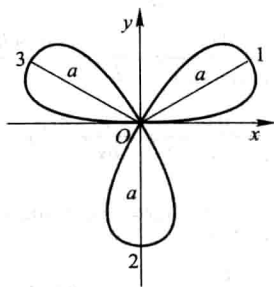
$$\rho^2 = a^2 \cos 2\theta$$

(15) 三叶玫瑰线



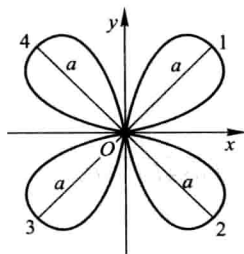
$$\rho = a \cos 3\theta$$

(16) 三叶玫瑰线



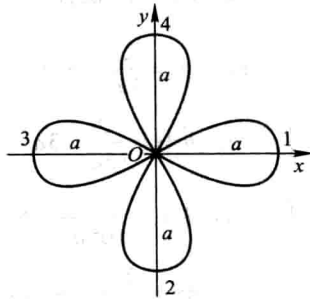
$$\rho = a \sin 3\theta$$

(17) 四叶玫瑰线



$$\rho = a \sin 2\theta$$

(18) 四叶玫瑰线



$$\rho = a \cos 2\theta$$